

Project Information Document

Source document for RAG ingestion — pi-system test

Project Title

AI-Powered Early Detection of Crop Disease Using Hyperspectral Imaging and Edge Computing

Principal Investigator

Dr. Meera Nair, Associate Professor, Department of Agricultural Engineering

Institution / Department

Indian Institute of Technology Bombay, Department of Agricultural and Food Engineering

Funding Source

Department of Biotechnology (DBT), Government of India — Grant No. DBT/PR45821/AGR/2024

Project Duration

April 2024 to March 2027 (3 years)

Research Objective

To develop a real-time, low-power crop disease detection system using hyperspectral cameras mounted on UAVs, processing data on edge devices using lightweight CNN models. The system aims to detect early-stage fungal and bacterial infections in wheat and rice crops with over 92% accuracy, enabling timely intervention and reducing pesticide usage by up to 40%.

Keywords

Hyperspectral Imaging, Edge Computing, Crop Disease Detection, UAV, CNN, Precision Agriculture, Deep Learning

Project Status

Active — Phase 1 completed (dataset collection). Phase 2 ongoing (model training and edge deployment).

Team Members

Dr. Meera Nair (PI), Dr. Suresh Iyer (Co-PI, CS Dept), Mr. Arjun Pillai (PhD Scholar), Ms. Priya Menon (MTech Student)

Total Budget

Rs. 87,50,000 (Eighty Seven Lakhs Fifty Thousand only)

Expected Outcomes

1. A validated hyperspectral dataset of 12 crop diseases across 3 crop varieties. 2. A lightweight CNN model achieving >92% accuracy at <50ms inference time on Jetson Nano. 3. Three peer-reviewed journal publications. 4. One provisional patent on the edge inference pipeline.